

Employee Attrition Prediction

IBM HR Analytics project - concepts, metrics, charts, and models explained from first principles

1,470

employees in dataset

237

employees left

16.12%

overall attrition rate

RF @ 0.25

final selected model

Prepared for Ruturaj Mokashi

Version 2 - rebuilt for readability, alignment, and concept depth

How this manual is designed

Not a superficial summary - this is a concept-building guide.

Reading promise

Each technical term is explained using the HR attrition project. You will see what the concept means, why it matters, how we calculated it, and how to interpret it for business.

Recommended reading style

Read in order. First understand the business question, then how KPIs are calculated, then charts, then preprocessing, then model evaluation, then model selection.

Part	What it explains	Why it matters
1. Business and data	Objective, target, train-test split	You need to know what the model is trying to predict.
2. KPIs and aggregation	groupby, counts, rates	This is how almost every EDA chart was created.
3. Charts	Box plots, violin plots, bars, lines	Charts answer business questions visually.
4. Preprocessing	Scaling and encoding	This turns raw data into model-readable data.
5. Evaluation	Confusion matrix, precision, recall, F1	This is how we judge if the model is useful.

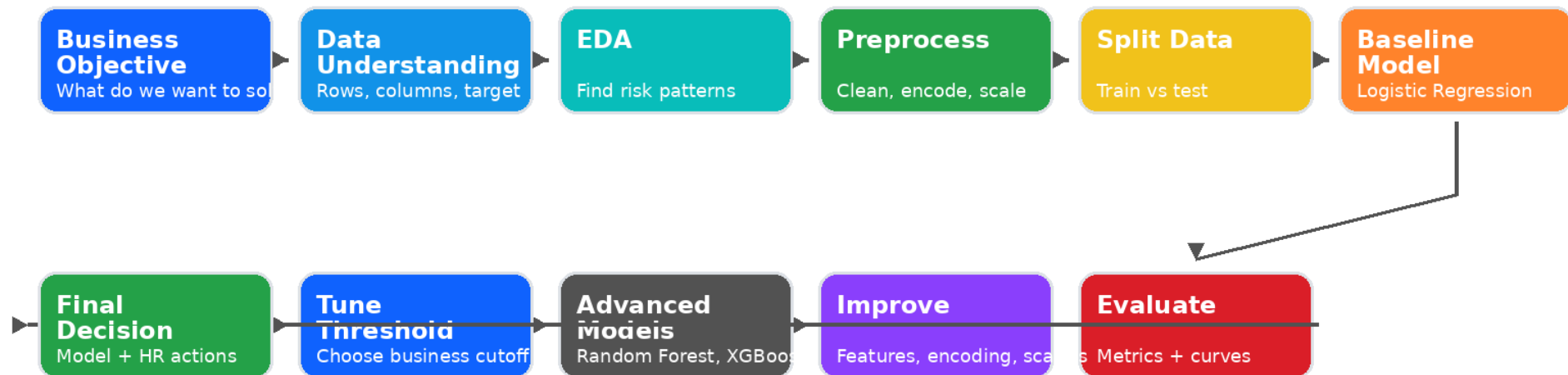
Important mindset

Machine learning is not magic. It is a structured process: define a business question, calculate signals, train on past examples, test on unseen examples, and choose the model that best supports the business objective.

Visual project flow

The full journey from business problem to final HR recommendation.

HR Attrition ML Workflow



Core idea: use past employee examples to learn patterns, then test on unseen employees as if they are future c

One key idea: the model did not see the test answers during prediction. The test set works like an exam answer sheet held by the teacher.

Section 1

Business Objective and Data

Before modeling, understand what the rows, columns, and target variable mean.

Business objective: predict attrition risk

The project is not about predicting for fun - it supports HR decision-making.

Business question

Which employees are more likely to leave, and what patterns are associated with attrition?

Machine learning question

Using employee features such as age, job role, income, overtime, satisfaction, and tenure, can we predict whether an employee will leave?

Target variable

Attrition
0 = Stayed
1 = Left

Prediction type

Binary classification. The model chooses between two outcomes: Stayed or Left.

Business use

Early-warning list for HR check-ins, not automatic employee decisions.

Why recall matters here

In attrition prediction, missing a real leaver can be costly. HR may prefer a model that catches more potential leavers, even if it creates some false alarms.

The data: what one row means

Each row is one employee. The target column says whether they left.

Employee	Age	JobRole	OverTime	Attrition
Employee A	41	Sales Executive	Yes	Left

Rows

1,470 employees. Each employee has features such as age, income, overtime, satisfaction, job role, and tenure.

Features X

All input columns used to make predictions.
Example: Age, MonthlyIncome, JobRole, OverTime.

Target y

The answer column: Attrition. This is what the model learns to predict.

1,233

employees stayed

237

employees left

16.12%

attrition rate

30

model input columns

Imbalanced dataset

Only 16.12% of employees left. This means accuracy can be misleading because a model can look good by predicting Stayed too often.

How prediction works when the data already exists

The model is tested as if the test answers are hidden.

1,029

training employees - model learns from these

441

test employees - model is examined on these

70 / 30

train-test split

Training set

The model sees features and the real Attrition answer. It learns patterns from these examples.

Test set

The model sees only features. The real Attrition answer is hidden until after prediction.

Exam analogy

Training data is practice questions with answers. Test data is the exam. The teacher has the answer sheet, but the student does not see it while answering. Evaluation means comparing the model answer with the hidden answer sheet.

Section 2

KPIs, GroupBy and Aggregation

This is how the EDA numbers and chart labels were calculated.

KPI = a business measurement

In this project, the most important KPI is attrition rate.

Attrition count

Number of employees who left in a group.
Example: 127 overtime employees left.

Employee count

Total employees in the group. Example: 416 employees worked overtime.

Attrition rate

Left count divided by total group count. Example:
 $127 / 416 = 30.5\%$.

$$\text{Attrition Rate} = \text{Employees Who Left} / \text{Total Employees in Group}$$

Example: Overtime = Yes -> $127 / 416 = 0.305 = 30.5\%$

How groupby works

1. Split the dataset into groups, such as OverTime = Yes and OverTime = No. 2. Count total employees in each group. 3. Count employees who left in each group. 4. Divide left count by total count. 5. Plot the result as a chart.

What groupby/aggregation did in our notebook

This is the engine behind most EDA charts.

OverTime	Stayed	Left	Total	Calculation	Attrition Rate
No	944	110	1054	110 / 1054	10.4%

Aggregation

Aggregation means compressing many rows into a summary. Instead of looking at 1,470 rows, we summarize by group: count, mean, median, min, max, or rate.

Why mean of Attrition works

After encoding Attrition as 0 = Stayed and 1 = Left, the average of Attrition becomes the attrition rate. Example: [1,0,0,1] has mean 0.5 = 50% left.

Did we use joins?

No major joins were needed because all columns were in one employee table. A join would be needed if employee details and department details were in separate tables and we had to merge them using a key like EmployeeNumber or DepartmentID.

How almost every EDA chart was built

1. Choose segment

Example: OverTime, JobRole, AgeBand, YearsAtCompanyBand

2. Group employees

Split rows into categories inside that segment.

3. Count employees

Total employees in each segment group.

4. Count leavers

Employees where Attrition = 1.

5. Calculate rate

$\text{attrition_rate} = \text{leavers} / \text{total employees}$.

6. Plot chart

Bar, dot, line, heatmap, or bubble depending on the question.

Core idea

Charts were not random decorations. Every chart was built to answer a business question using a KPI, usually attrition rate by group.

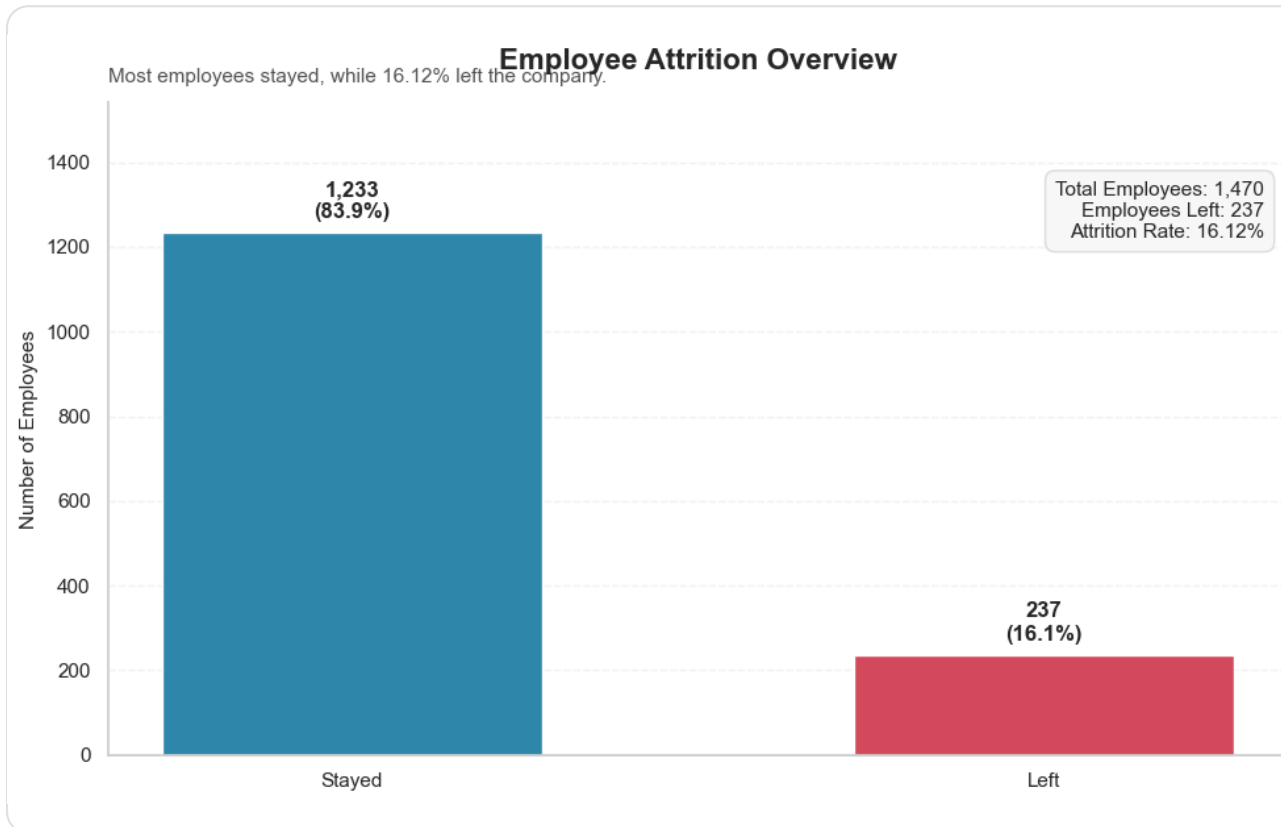
Section 3

Charts and Visual Thinking

Charts are business questions converted into visual evidence.

Chart: Employee attrition overview

Question: How many employees stayed vs left?



How we built it

Count employees by Attrition status. Add percentage labels and total employee count.

How to read it

The Stayed bar is much larger than the Left bar. This confirms class imbalance.

Business takeaway

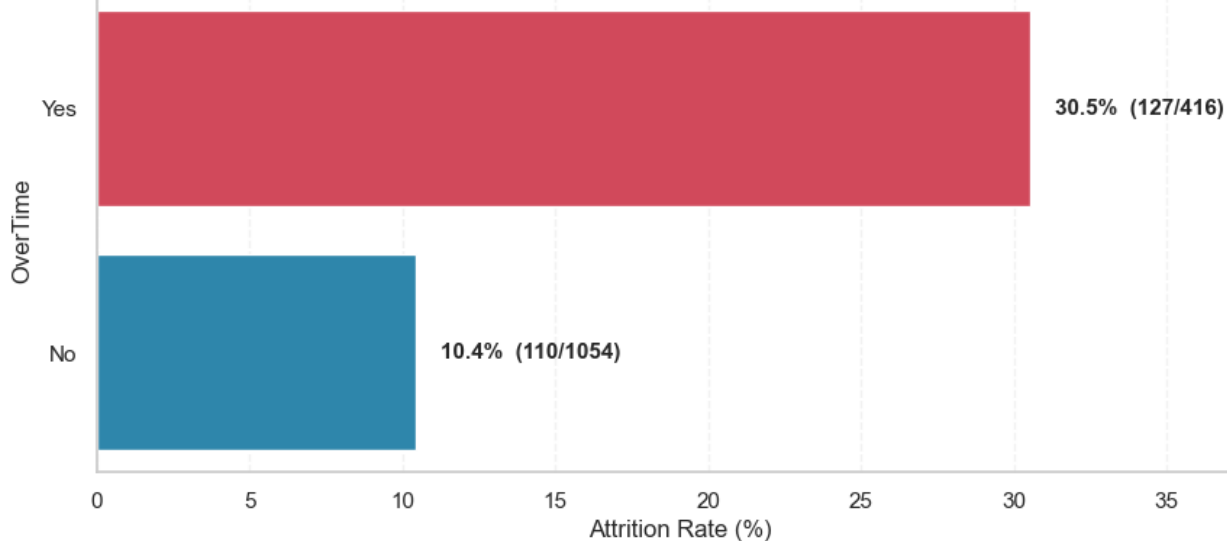
Most employees stayed, but 237 employees left. The baseline attrition rate is 16.12%.

Chart: Attrition by overtime

Question: Does overtime relate to attrition?

Attrition Rate by Overtime Status

Employees working overtime show a higher attrition rate.



How we built it

Group by OverTime, count leavers and total employees, then calculate attrition rate.

How to read it

OverTime = Yes has a much longer bar than OverTime = No.

Business takeaway

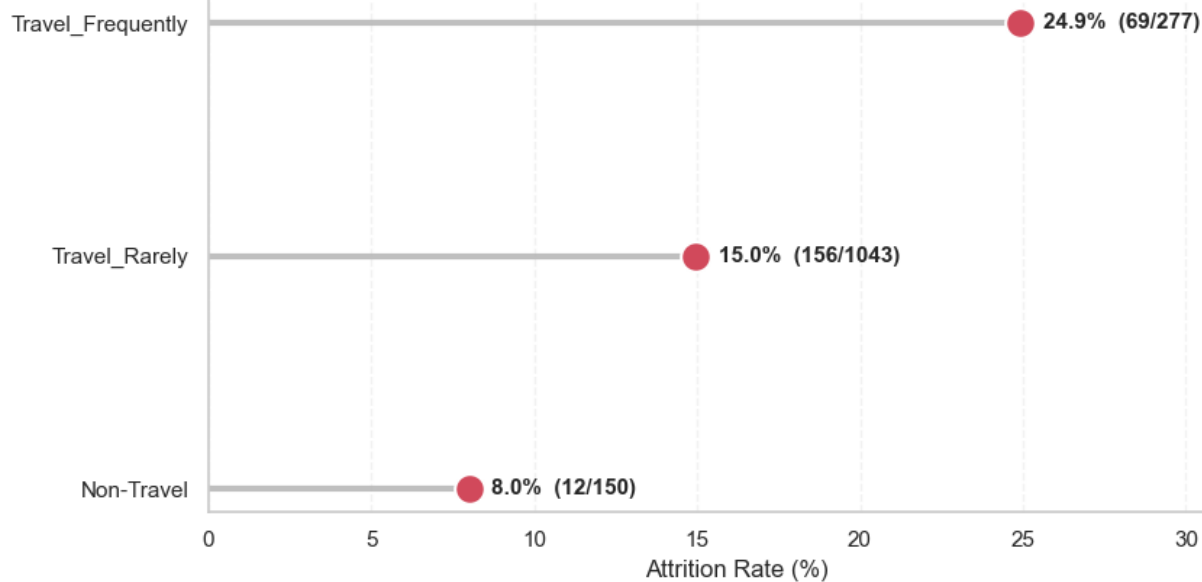
Employees working overtime are a high-risk group. This became one of the strongest model signals.

Chart: Attrition by business travel

Question: Does frequent travel increase risk?

Attrition Rate by Business Travel Frequency

Attrition rate varies across travel frequency groups.



How we built it

Group by BusinessTravel and calculate attrition rate per category.

How to read it

Travel_Frequently is highest, Non-Travel is lowest.

Business takeaway

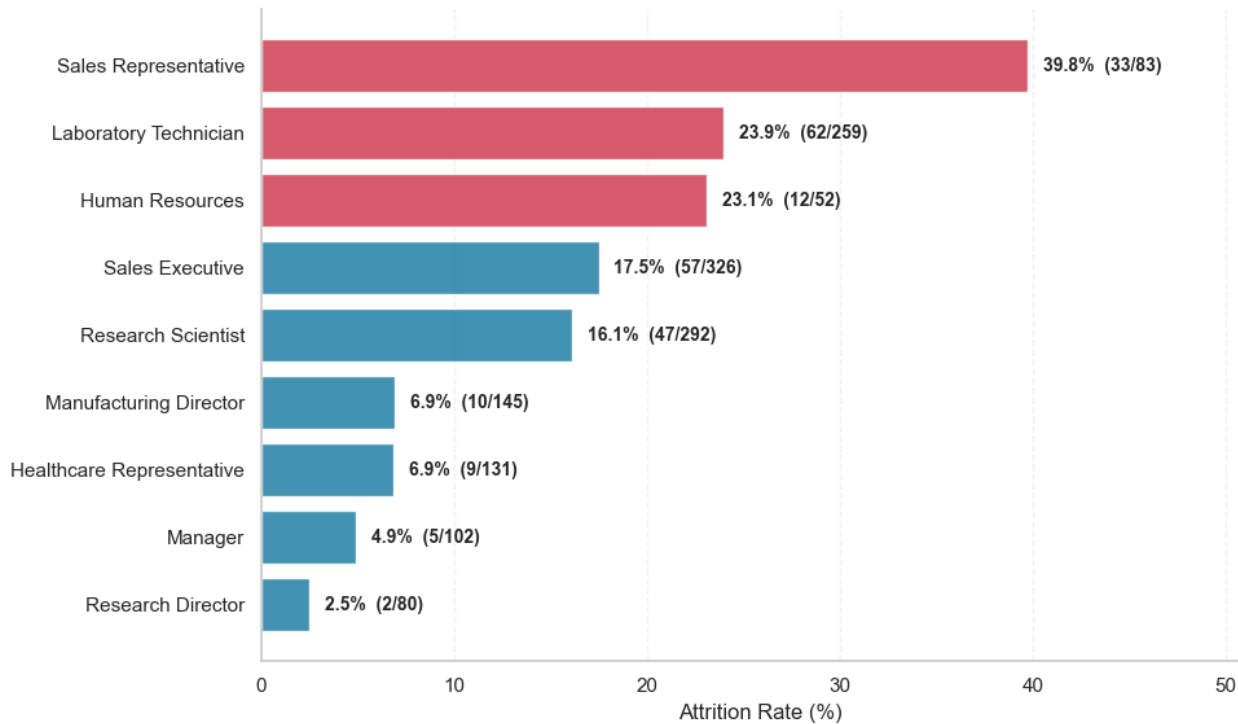
Frequent business travel is associated with higher attrition risk.

Chart: Attrition by job role

Question: Which roles are highest risk?

Attrition Rate by Job Role

Roles highlighted in red are in the top quartile of attrition risk.



How we built it

Group by JobRole, calculate attrition rate, sort from high to low, and highlight top-risk roles.

How to read it

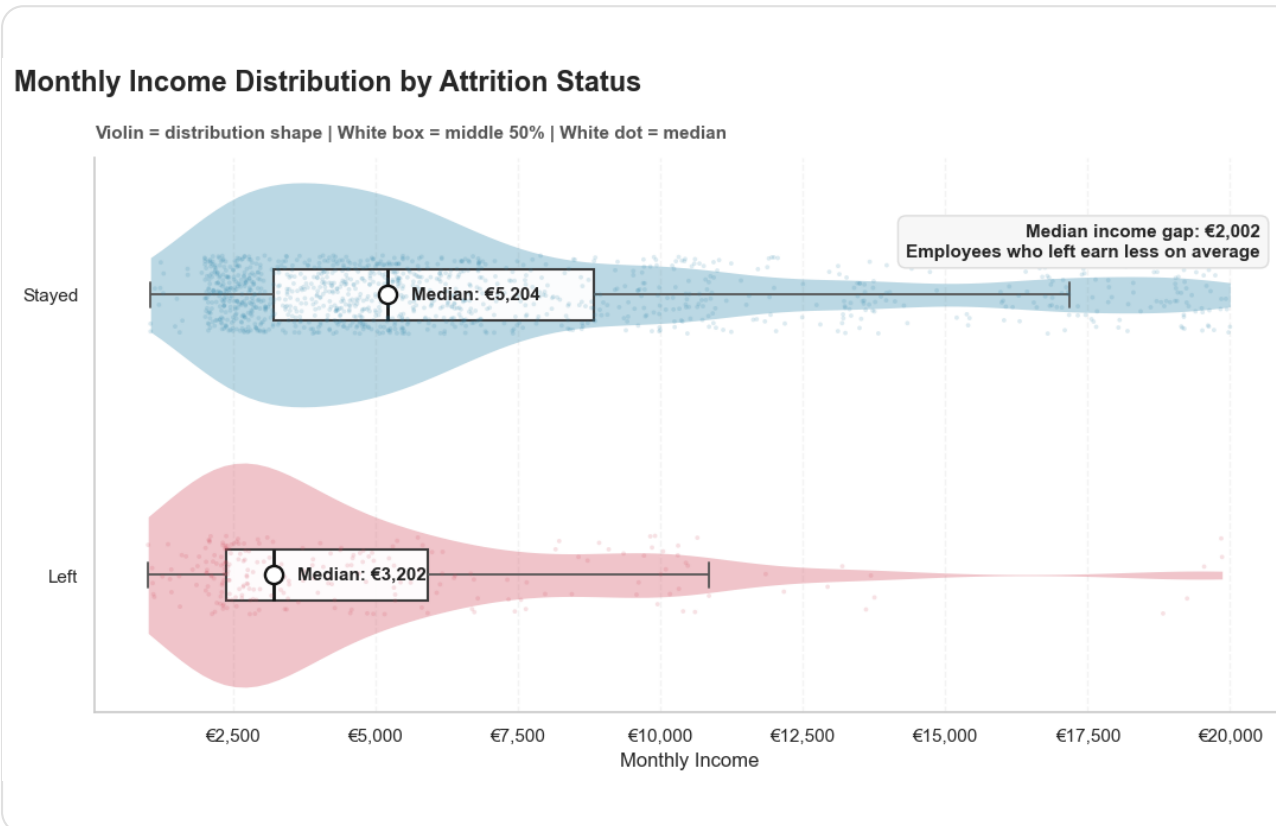
Sales Representative is clearly the highest-risk role.

Business takeaway

Job role matters. Sales Representative and Laboratory Technician require attention.

Chart: Income distribution by attrition

Question: Do leavers earn differently from stayers?



How we built it

Plot MonthlyIncome distribution for Stayed vs Left using violin shape, box, median dot, and individual points.

How to read it

The Left group is shifted toward lower income, and its median is lower.

Business takeaway

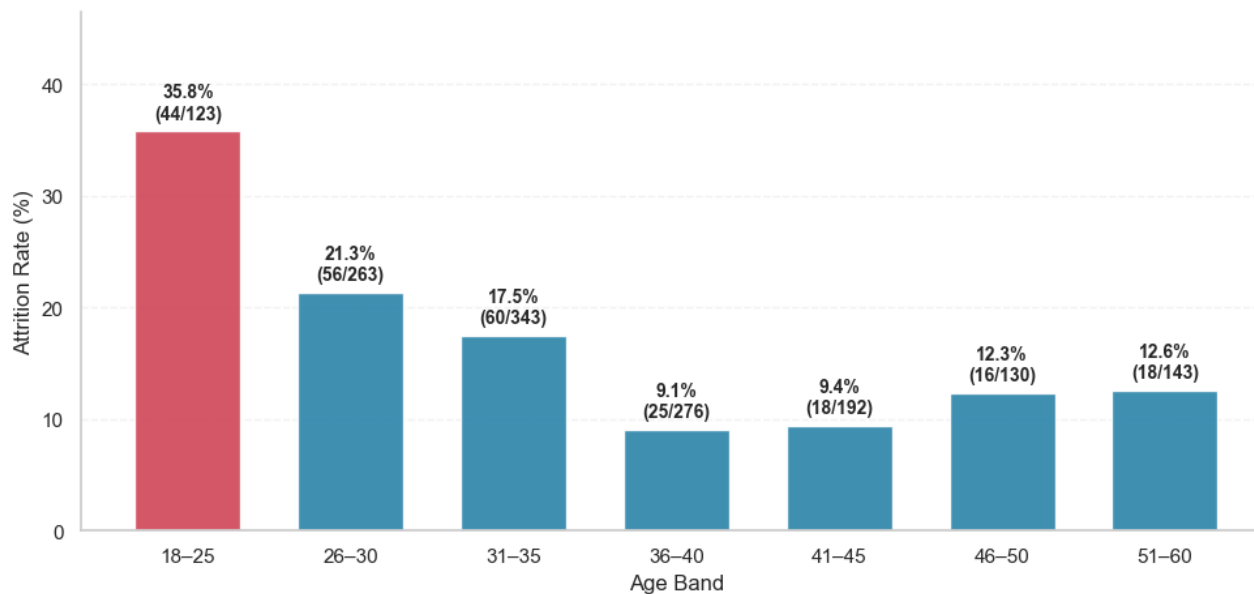
Lower monthly income is associated with higher attrition risk.

Chart: Attrition by age band

Question: Are younger employees more likely to leave?

Attrition Rate by Age Band

Younger employees show higher attrition rates in this dataset.



How we built it

Create age bands, group by band, calculate attrition rate.

How to read it

The 18-25 group is much higher than older groups.

Business takeaway

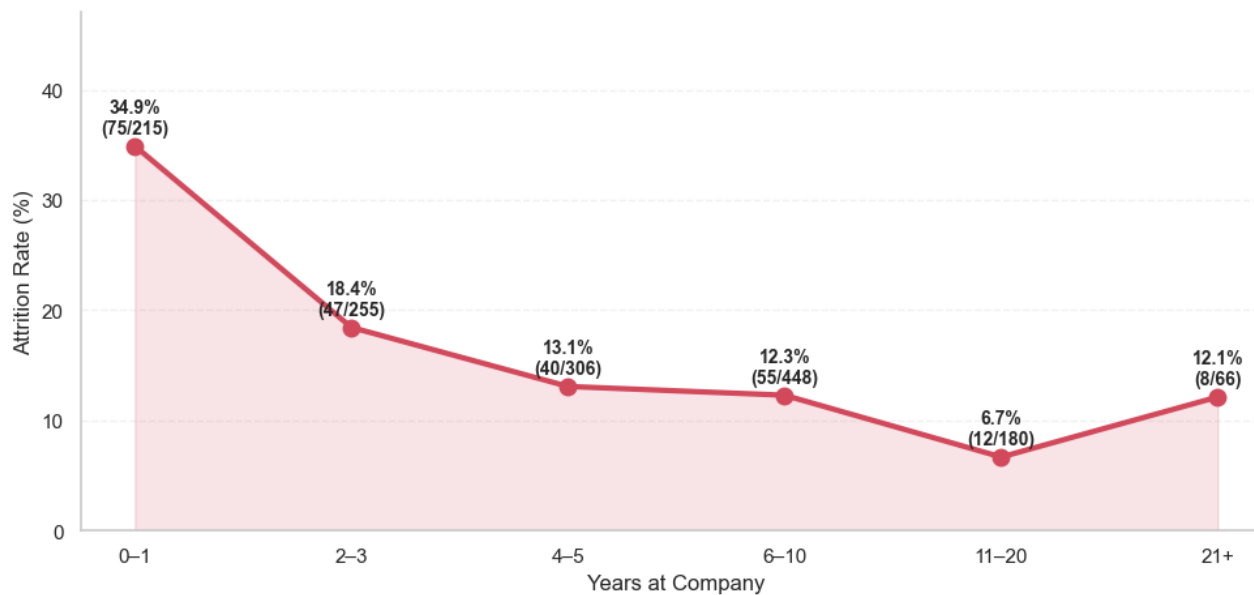
Younger employees are a major high-risk segment.

Chart: Attrition by years at company

Question: Is early tenure risky?

Attrition Rate by Years at Company

Early-tenure employees show higher attrition risk.



How we built it

Create tenure bands and calculate attrition rate by group.

How to read it

0-1 years has the highest attrition rate, then risk drops.

Business takeaway

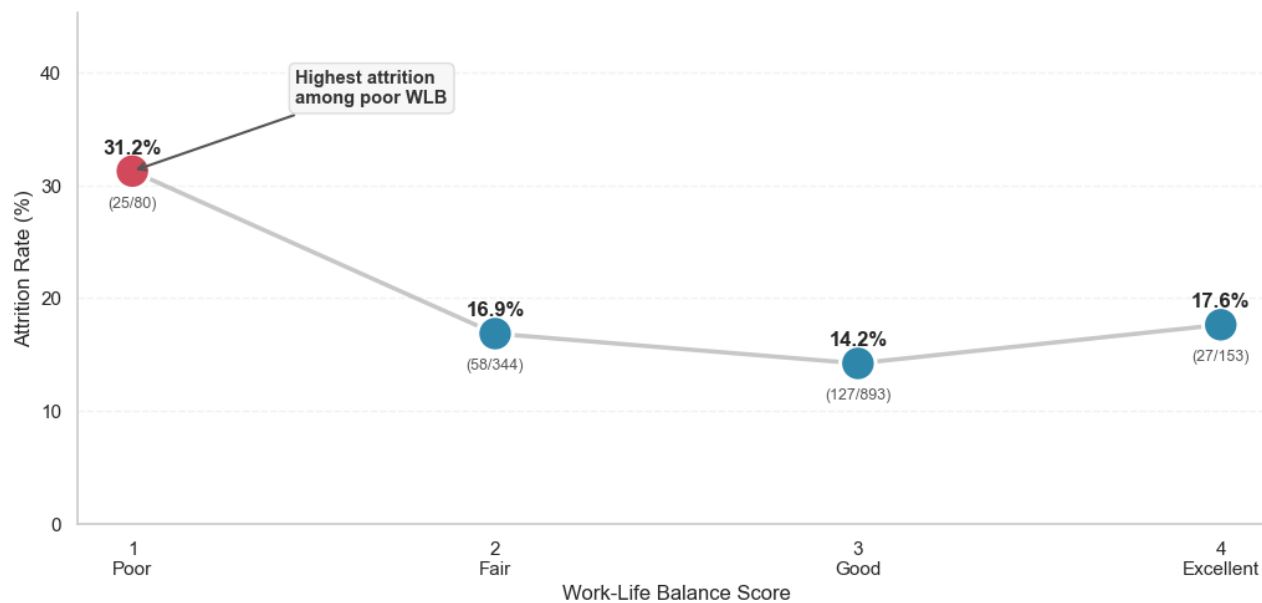
Early-tenure employees need stronger onboarding and first-year support.

Chart: Work-life balance risk

Question: Does poor work-life balance matter?

Attrition Rate by Work-Life Balance

Poor work-life balance stands out as the highest-risk group; other scores are mixed.



How we built it

Group by WorkLifeBalance score and calculate attrition rate.

How to read it

Poor work-life balance has the highest risk.

Business takeaway

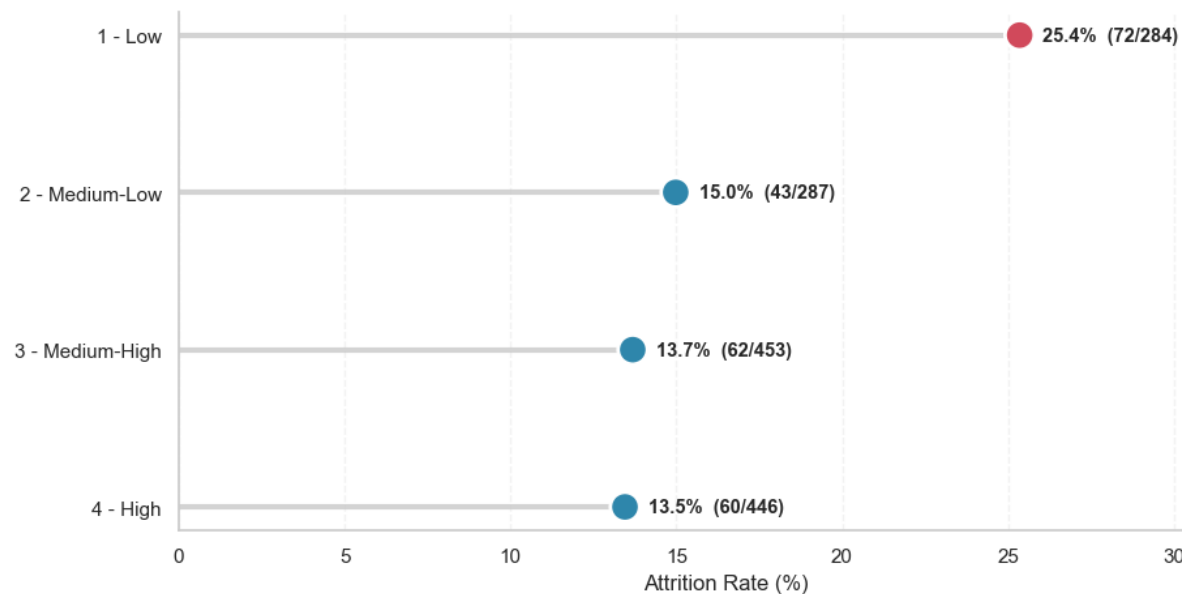
Work-life balance is an actionable HR area.

Chart: Environment satisfaction

Question: Does low environment satisfaction matter?

Attrition Rate by Environment Satisfaction

Lower environment satisfaction groups show higher attrition rates.



How we built it

Group by EnvironmentSatisfaction score and calculate attrition rate.

How to read it

Low environment satisfaction has the highest rate.

Business takeaway

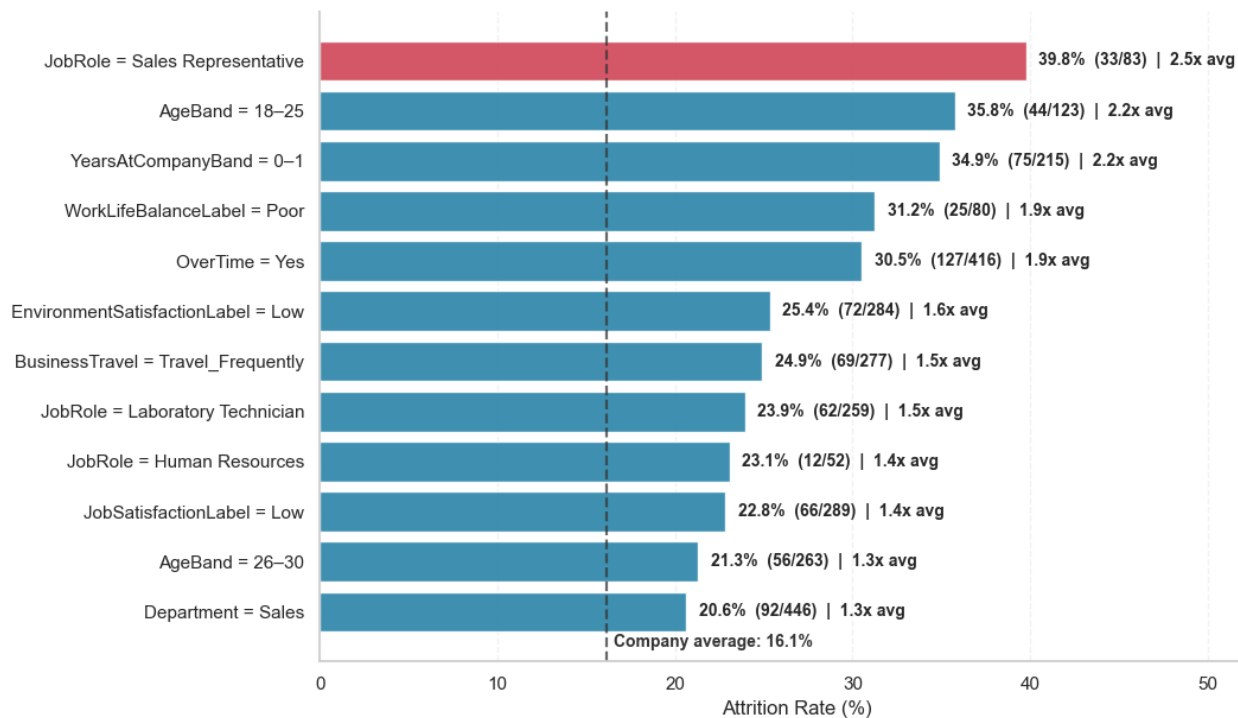
Workplace environment and manager support may affect retention.

Chart: Top risk segments

Question: Which segments should HR focus on first?

Top Attrition Risk Segments

Only groups with at least 50 employees are included to avoid tiny-group noise.



How we built it

Combine several grouped attrition-rate summaries and keep groups with enough employees to avoid tiny-group noise.

How to read it

Top segments are above the company average line.

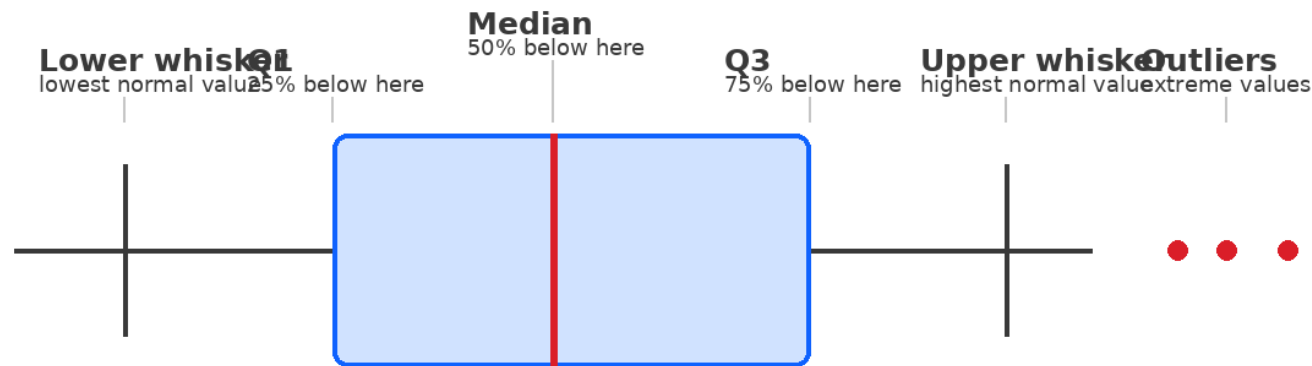
Business takeaway

The highest-risk groups are Sales Representatives, young employees, early-tenure employees, poor WLB, overtime, and low environment satisfaction.

Why the income chart looked mysterious

Box plots and violin plots show distribution, not just averages.

Box Plot Anatomy: What the Mysterious Box Means



How to read it in our income chart

- The box is the middle 50% of employees. It shows the normal income band.
- The red/black line inside the box is the median income, not the average.
- Whiskers show the normal spread outside the middle 50%.
- Dots beyond whiskers are outliers: unusually high or low values.
- If the Left group box is shifted lower than Stayed, leavers generally earn less.

Box plot vs violin plot: when to use each

Both explain distribution, but they emphasize different things.

Box plot

Shows median, middle 50%, whiskers, and outliers. It is compact and good for comparing groups, but it hides the exact distribution shape.

Violin plot

Shows the distribution shape. Wide parts mean many employees are concentrated there. It is more visual, but can look unfamiliar at first.

Chart element	What it means	In our income chart
White dot	Median	Middle employee income in each group

How to interpret our income chart

The Left group has a lower median income than the Stayed group. This does not prove salary causes attrition, but it tells us income is related to attrition patterns and may be useful for the model.

Section 4

Preprocessing: Preparing Data for Models

Models need numeric, consistent, leakage-safe input.

Why preprocessing was needed

Machine learning models do not understand raw business text directly.

Numeric features

Already numbers: Age, MonthlyIncome, DistanceFromHome, YearsAtCompany, JobSatisfaction. But they can be on very different scales.

Categorical features

Text categories: BusinessTravel, Department, JobRole, MaritalStatus, OverTime. These must be converted into numbers.

Raw value	Problem	Preprocessing solution
MonthlyIncome = 10,000	Large numeric scale	Scaling
JobSatisfaction = 8	Small numeric scale	Scaling

Pipeline mindset

Preprocessing was placed inside sklearn pipelines so the same rules are learned from the training data and then applied safely to the test data. This reduces manual mistakes and data leakage.

StandardScaler, MinMaxScaler, RobustScaler

Same information, different numerical measurement scale.

Scaling: Same Meaning, Different Measurement Scale

StandardScaler

Center around average

Formula: $(\text{value} - \text{mean}) / \text{std}$

Best mental model: How far from average?

MinMaxScaler

Squeeze into 0 to 1

Formula: $(\text{value} - \text{min}) / (\text{max} - \text{min})$

Best mental model: Percentage-like position.

RobustScaler

Use median and middle 50%

Formula: $(\text{value} - \text{median}) / \text{IQR}$

Best mental model: Less disturbed by outliers.

Why we needed scaling in this project

Feature	Raw range	Problem
MonthlyIncome	1009 to 19973	Very large numbers
JobSatisfaction	1 to 4	Very small numbers
Age	18 to 60	Medium scale

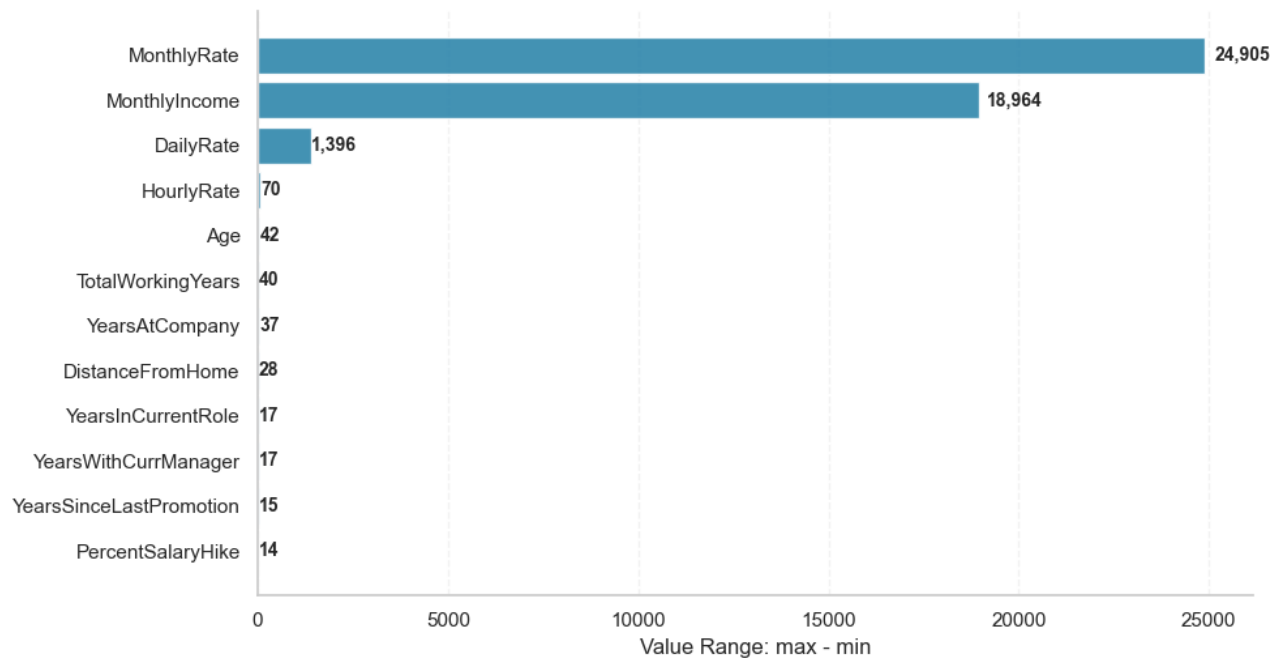
Scaling prevents large-number columns from dominating models such as Logistic Regression.

Why scaling was necessary

Some features had ranges in the thousands; others ranged only from 1 to 4.

Numeric Feature Ranges Before Scaling

Large differences in feature ranges show why scaling is needed.



Problem

MonthlyRate and MonthlyIncome have huge ranges, while satisfaction scores are tiny. Some models can be affected by these different scales.

Solution

Scaling transforms numeric columns so they become more comparable for models such as Logistic Regression.

Important

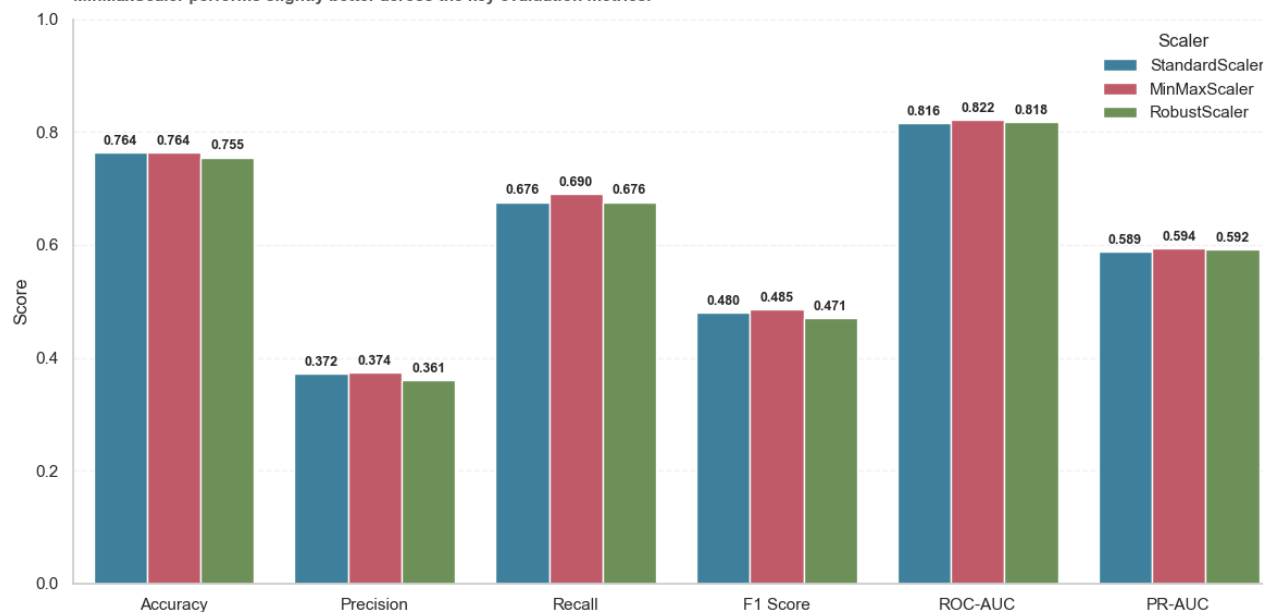
Scaling does not change business meaning. High salary is still high salary; it is just represented differently.

How scalers changed model performance

We tested preprocessing choices instead of assuming one scaler is best.

Scaler Comparison for Logistic Regression

MinMaxScaler performs slightly better across the key evaluation metrics.



Best scaler

MinMaxScaler performed slightly better than StandardScaler and RobustScaler for Logistic Regression.

Why not huge difference?

All scalers preserve the same information. They mostly change how values are represented numerically.

Practical lesson

Preprocessing is part of modeling. Small choices like scaler type can change recall, F1, ROC-AUC, and PR-AUC. Test them objectively.

OneHotEncoder and TargetEncoder

Two different ways to convert categories into numbers.

OneHotEncoder

Creates separate 0/1 columns for each category. Example: OverTime_Yes = 1, OverTime_No = 0. It avoids fake ranking between categories.

TargetEncoder

Replaces each category with target-based information, such as average attrition risk for that category. Example: Sales Representative -> high risk value.

Encoding method	Strength	Risk
OneHotEncoder	Simple and safe	Can create many columns
TargetEncoder	Creates less columns	Can leak test information if not used properly

Data leakage warning

TargetEncoder uses the target column. If fitted on the full dataset before train-test split, the model indirectly sees test answers. That is why it must be fitted only on training data, preferably inside a pipeline.

Section 5

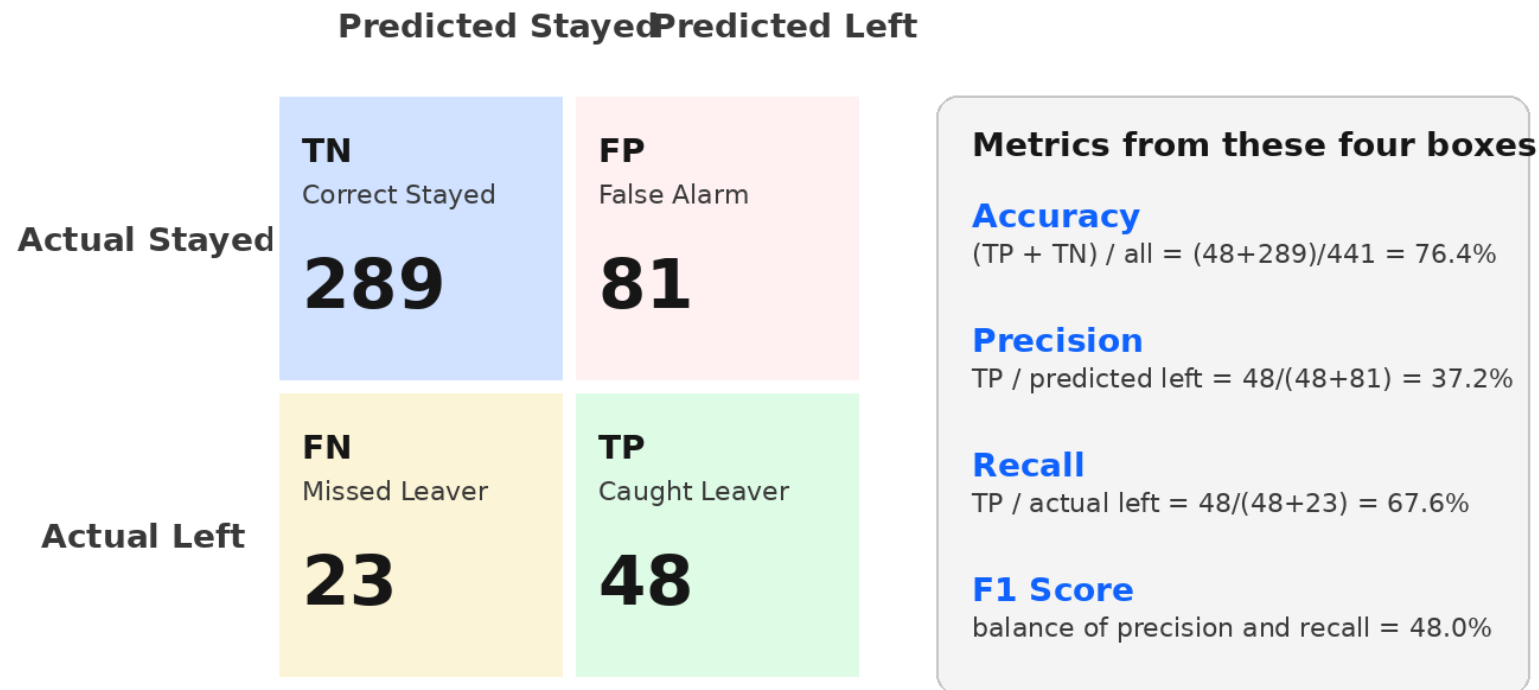
Model Evaluation Without Gibberish

This is the core section: how to know whether a model is useful.

Confusion matrix: the model report card

Every classification metric starts from four boxes.

Confusion Matrix: How Model Predictions Are Checked



For HR attrition, the dangerous error is FN: employees who actually left but the model predicted they stayed

Our baseline confusion matrix

This is how the Logistic Regression baseline made correct and wrong predictions.

Confusion Matrix — Baseline Logistic Regression

Shows correct predictions and model errors on the test set.

	Predicted	
	Stayed	Left
Actual Stayed	289	81
Actual Left	23	48

Correct Stayed: 289

Employees who actually stayed and were predicted as stayed. This is good.

False Alarms: 81

Employees predicted as leavers but actually stayed. HR may check in unnecessarily.

Missed Leavers: 23

Employees predicted as stayed but actually left. This is the dangerous HR error.

Caught Leavers: 48

Employees who actually left and were correctly flagged as leavers.

The four basic model metrics

Each metric answers a different question.

Metric	Question it answers	Formula using boxes	Our baseline
Accuracy	How often was the model correct overall?	$(TP + TN) / \text{all}$	76.4%
Precision	When model says Left, how often is it correct?	$TP / (TP + FP)$	37.2%
Recall	Of all real leavers, how many did it catch?	$TP / (TP + FN)$	67.6%
F1 Score	Balance between precision and recall	$2PR / (P + R)$	48.0%

Why accuracy misleads here

Because most employees stayed, a model can get high accuracy by mostly predicting Stayed. That may be useless for detecting leavers.

Why recall matters

For HR attrition, missing actual leavers is costly. Recall tells us how many real leavers the model caught.

Plain English summary

Precision controls false alarms. Recall controls missed leavers. F1 balances both. For our business goal, recall and F1 are more useful than accuracy alone.

Why you cannot maximize everything at once

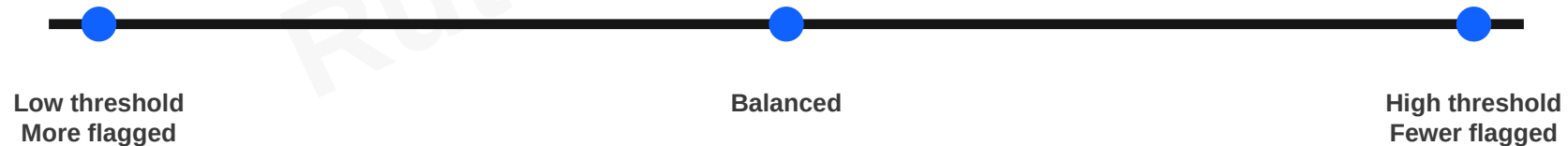
A model threshold controls how aggressive the model is.

High recall setting

The model flags more employees. It catches more true leavers, but also creates more false alarms. This is useful when missing leavers is expensive.

High precision setting

The model flags fewer employees. When it flags someone, it is more often correct, but it may miss many real leavers.



In our final model

Random Forest worked poorly at the default threshold 0.50, but after lowering the threshold to 0.25, recall and F1-score became much better for the HR use case.

AUC metrics: ranking quality, not one fixed decision

These metrics use predicted probabilities, not only final class labels.

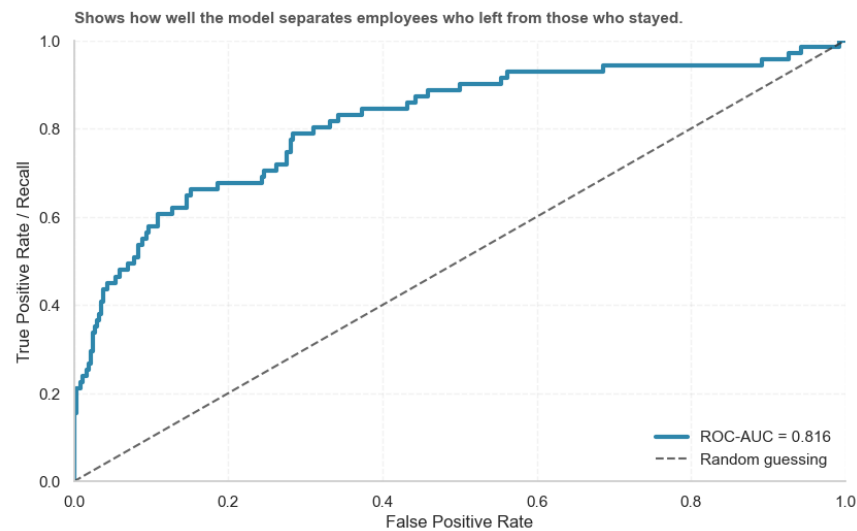
ROC-AUC

Measures how well the model separates leavers from stayers across thresholds. 0.50 is random guessing. Higher is better.

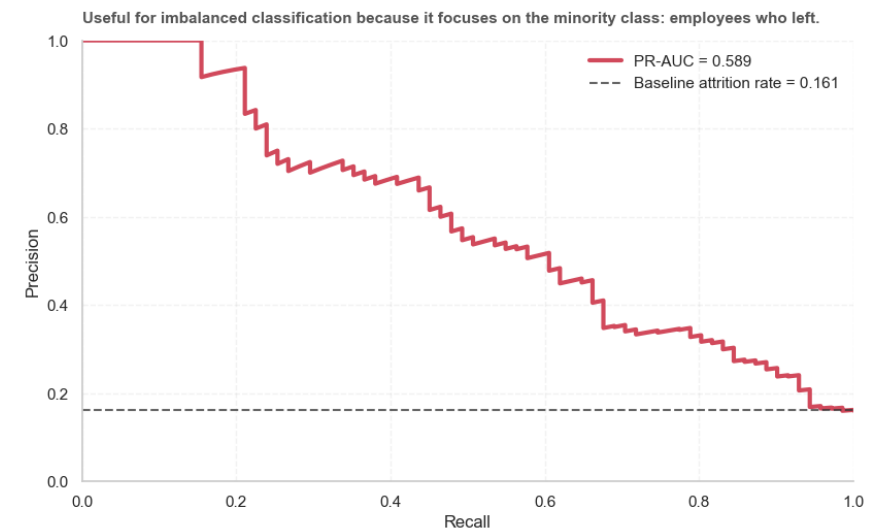
PR-AUC

Focuses on the minority class. It is often more useful than ROC-AUC when the positive class is rare, as in attrition.

ROC Curve — Baseline Model



Precision-Recall Curve — Baseline Model



Key idea

ROC-AUC and PR-AUC judge probability ranking. Changing the threshold changes precision/recall/F1, but not the underlying ranking curve.

Section 6

Models: What They Are Doing

No black boxes: each model is explained using our attrition project.

Logistic Regression: a weighted risk score converted into probability

Despite the name, it is used for classification.

What it does

It learns a weight for each feature. Positive weights increase attrition probability; negative weights reduce attrition probability.

Example idea

OverTime_Yes and Sales Representative received positive coefficients, so they push predictions toward higher attrition risk.

$$\text{Risk score} = \text{intercept} + \text{weight1} * \text{feature1} + \text{weight2} * \text{feature2} + \dots$$

Then the risk score is converted into a probability between 0 and 1.

Why we used it

Good baseline, explainable coefficients, works well with scaled numeric data and one-hot encoded categories.

Limitation

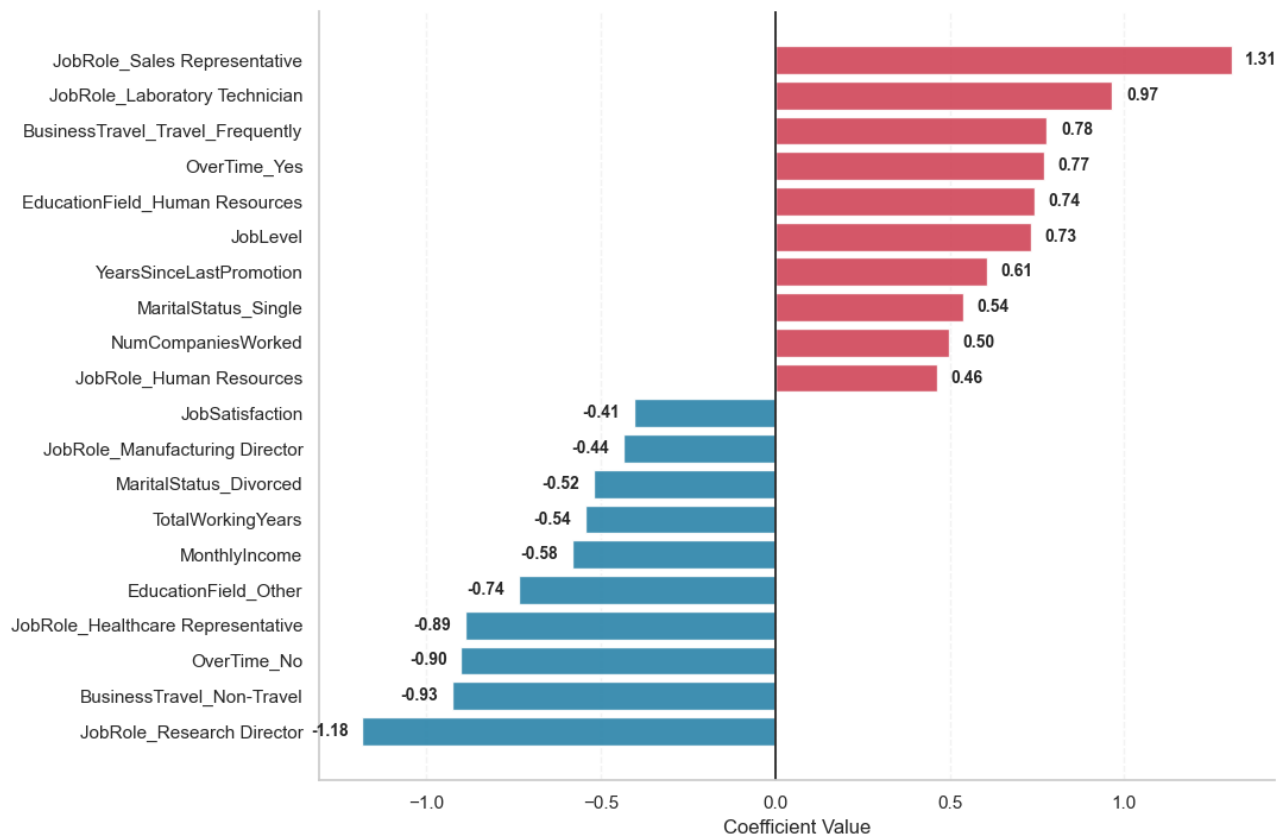
It may miss complex non-linear patterns because it is mostly a linear model.

Model interpretation: coefficients

This chart explains which features push predictions up or down.

Top Logistic Regression Coefficients

Positive coefficients increase predicted attrition risk; negative coefficients reduce it.



Positive side

Increases predicted attrition risk. Sales Representative, Laboratory Technician, Travel Frequently, and OverTime Yes were strong risk signals.

Negative side

Reduces predicted attrition risk. Research Director, Non-Travel, OverTime No, higher income, and more working years reduce risk.

Caution

Coefficients show association learned by the model. They do not prove direct cause.

Random Forest: many decision trees voting together

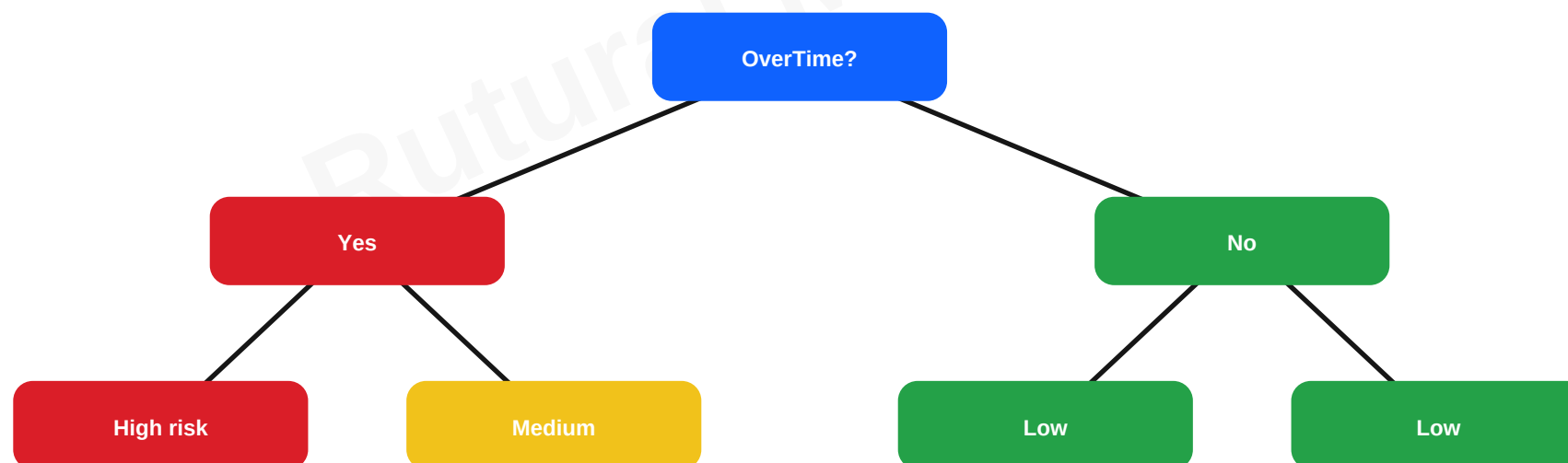
A tree-based model that captures more complex patterns.

Decision tree idea

A tree asks a sequence of questions: Is OverTime Yes? Is Age below 25? Is MonthlyIncome low? At the end, it predicts Stay or Leave.

Forest idea

A Random Forest builds many trees using different samples and features. The final probability comes from many trees voting.



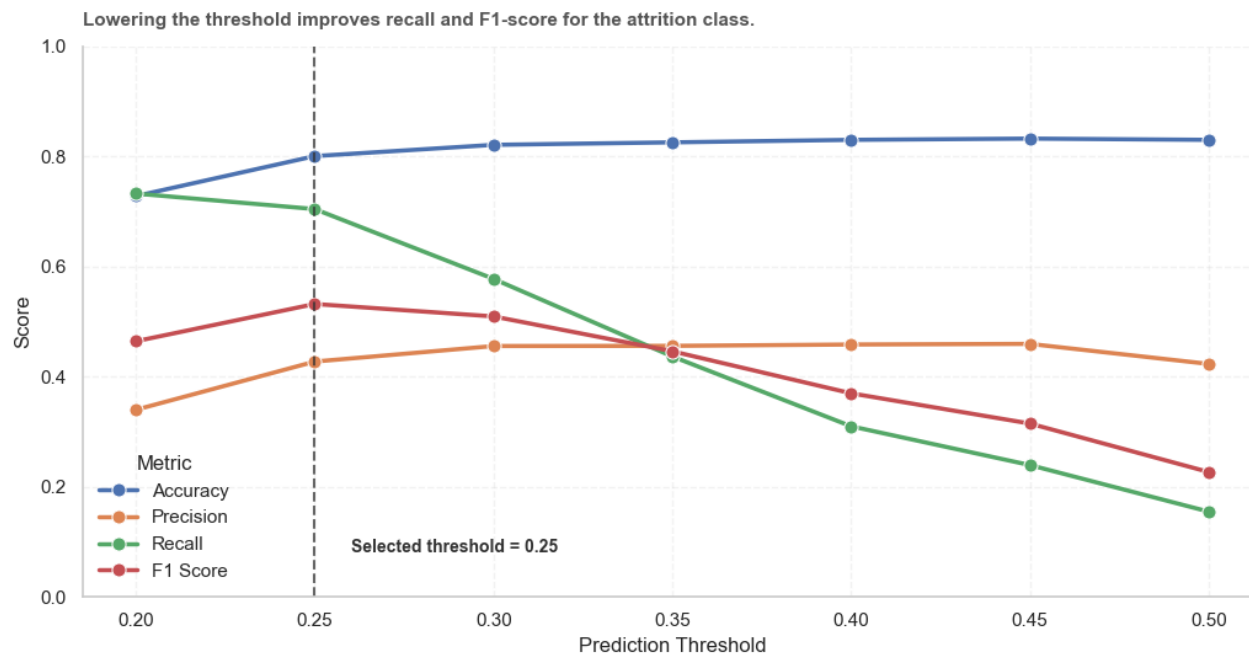
Why threshold tuning mattered

At threshold 0.50, Random Forest was too conservative. At threshold 0.25, it caught many more actual leavers and became the final selected model.

Threshold tuning: making Random Forest useful for HR

The same probabilities can be converted into decisions at different cutoffs.

Random Forest Threshold Tuning



Default threshold 0.50

Too strict: recall was only 15.5%, so most actual leavers were missed.

Selected threshold 0.25

Better HR balance: recall became 70.4% and F1-score became 53.2%.

Business meaning

Flag employees with at least 25% predicted risk. This creates more HR follow-up cases but catches more real leavers.

XGBoost: trees that learn from previous mistakes

A strong industry model, but not always best for every business goal.

What it does

XGBoost builds trees sequentially. Each new tree tries to correct errors made by earlier trees. This is called boosting.

Why it is popular

Often strong on structured/tabular data. Used widely in competitions and industry when tuned carefully.

Our result

Accuracy and precision were reasonable, but recall was lower than tuned Random Forest and Logistic Regression.

Lesson

Advanced models do not automatically win. Model choice depends on metrics and business objective.

Project comparison

LogReg + MinMaxScaler: recall 69.0%, F1 48.5%. Random Forest threshold 0.25: recall 70.4%, F1 53.2%. XGBoost: recall 45.1%, F1 44.8%. Therefore XGBoost was not selected.

Models used in industry: where they fit

A quick map of common model families.

Model family	Common use	Strength	Limitation
Logistic Regression	Risk scoring, credit, churn baseline	Interpretable and fast	Limited complexity
Random Forest	Tabular classification/regression	Handles non-linear patterns	Can be less interpretable
XGBoost / LightGBM	High-performance tabular ML	Very strong predictive power	Needs tuning
CatBoost	Categorical-heavy tabular data	Handles categories well	Another library dependency
Neural Networks	Images, text, complex data	Flexible and powerful	Needs more data and tuning
LLMs	Text-heavy HR signals, documents	Understands language context	Privacy, cost, governance issues

Professional takeaway

Start simple and interpretable. Build a baseline. Then test stronger models. Do not choose a model because it sounds advanced; choose it because it improves the business metric that matters.

Section 7

Final Model and Business Decision

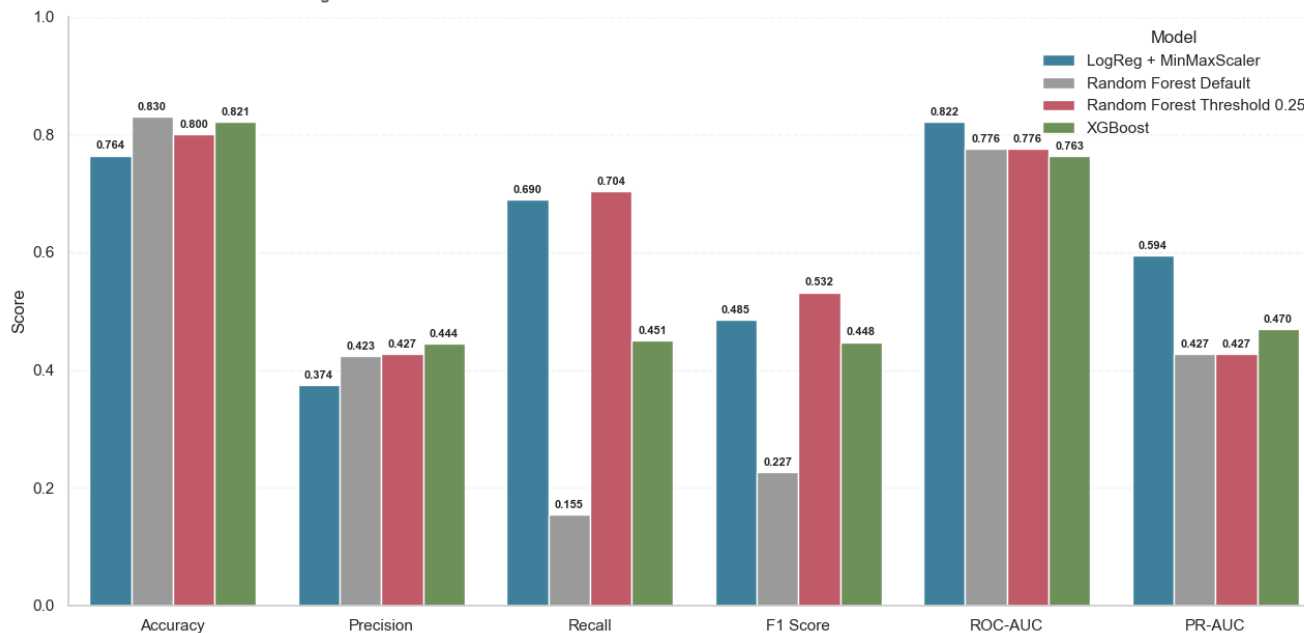
The final model was selected based on the HR objective, not accuracy alone.

Final model comparison

This is where the business decision becomes clear.

Final Model Comparison

The tuned Random Forest model gives the best recall and F1-score for the HR attrition use case.



Best ranking

LogReg + MinMaxScaler had the best ROC-AUC and PR-AUC.

Highest accuracy

Random Forest default had high accuracy but terrible recall. It missed most leavers.

Final choice

Random Forest at threshold 0.25 had the best recall and F1-score, making it best for HR early warning.

Final selected model: Random Forest with threshold 0.25

Selected because it best supports the HR early-warning objective.

80.0%

Accuracy

42.7%

Precision

70.4%

Recall

53.2%

F1 Score

77.6%

ROC-AUC

42.7%

PR-AUC

Why this model won

It catches the highest proportion of actual leavers and gives the best balance between precision and recall.

How HR should use it

Use predictions as an early-warning signal for supportive actions: check-ins, workload review, mentoring, and retention support.

Ethics

Do not use predictions to punish employees. Use them to start supportive conversations.

From model insight to HR action

The model is useful only if it leads to better decisions.

Risk area	What the analysis showed	Possible HR action
Overtime	Overtime employees had much higher attrition	Review workload and staffing
Early tenure	0-1 year employees had high attrition	Improve onboarding and first-year support
Sales Representatives	Highest job-role attrition	Review targets, incentives, and support
Young employees	18-25 group was high risk	Mentoring and career pathing
Work-life balance	Poor WLB had high attrition	Team-level interventions
Income / growth	Leavers tended to have lower income	Review pay and progression

Final business interpretation

Attrition risk is concentrated in specific segments. The model helps prioritize attention, but HR must combine model output with human judgment, manager context, and ethical safeguards.

What you should now understand

Use this as your foundation checklist.

- ✓ A row is one employee; the target is Attrition.
- ✓ groupby creates segment-level KPIs such as attrition by job role or overtime.
- ✓ OneHotEncoder converts categories into 0/1 columns.
- ✓ A confusion matrix contains four boxes: TN, FP, FN, TP.
- ✓ Recall tells us how many actual leavers we caught.
- ✓ F1 balances precision and recall.
- ✓ The final model is chosen based on the business objective, not only accuracy.
- ✓ Attrition rate is leavers divided by total employees in a group.
- ✓ Scaling changes numeric representation, not business meaning.
- ✓ TargetEncoder can be useful but can leak target information if used incorrectly.
- ✓ Accuracy can mislead when the dataset is imbalanced.
- ✓ Precision tells us how many predicted leavers were truly leavers.
- ✓ Threshold tuning changes how aggressive the model is.

Sources and references

Technical definitions were cross-checked against official documentation.

ID	Source	URL / note
S1	scikit-learn StandardScaler documentation	https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.StandardScaler.html
S2	scikit-learn MinMaxScaler documentation	https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.MinMaxScaler.html
S3	scikit-learn RobustScaler documentation	https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.RobustScaler.html
S4	scikit-learn confusion_matrix documentation	https://scikit-learn.org/stable/modules/generated/sklearn.metrics.confusion_matrix.html
S5	scikit-learn precision_score documentation	https://scikit-learn.org/stable/modules/generated/sklearn.metrics.precision_score.html
S6	scikit-learn recall_score documentation	https://scikit-learn.org/stable/modules/generated/sklearn.metrics.recall_score.html
S7	scikit-learn f1_score documentation	https://scikit-learn.org/stable/modules/generated/sklearn.metrics.f1_score.html
S8	scikit-learn roc_auc_score documentation	https://scikit-learn.org/stable/modules/generated/sklearn.metrics.roc_auc_score.html
S9	XGBoost Python API documentation	https://xgboost.readthedocs.io/
S10	Category Encoders TargetEncoder documentation	https://contrib.scikit-learn.org/category_encoders/targetencoder.html
S11	Project notebook outputs and charts	Raturaj Mokashi IBM HR Attrition notebook, generated during this project

Citation note

The dataset calculations, charts, and model metrics in this manual come from the project notebook outputs. Technical definitions and API meanings were checked against the official documentation listed above.